

MSDS 7333: Case Study 4
Ensemble Methods for Corporate Bankruptcy
Prediction

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Chapter 1

Introduction

Corporate bankruptcy prediction has emerged as a critical area of research in financial analytics, offering substantial value to stakeholders including investors, creditors, and policymakers. The ability to accurately predict bankruptcy enables informed decision-making, risk assessment, and proactive financial management strategies. This study leverages a comprehensive bankruptcy dataset spanning multiple time horizons to develop robust ensemble models for bankruptcy prediction.

The dataset utilized in this research comprises bankruptcy information for companies across five different time periods (from 1 to 5-year horizons). Each temporal dataset contains 64 financial features that capture various aspects of company performance, including liquidity ratios, profitability metrics, leverage indicators, and operational efficiency measures. The complete dataset encompasses thousands of companies, providing a rich foundation for developing predictive models.

The financial features in the dataset represent key financial ratios and indicators commonly used in financial analysis and bankruptcy prediction. These include metrics such as working capital ratios, debt-to-equity ratios, return on assets, current ratios, and various profitability measures. The comprehensive nature of these features allows for a holistic assessment of company financial health across multiple dimensions.

The primary objective of this research is to develop and compare ensemble machine learning models for bankruptcy prediction using the consolidated multi-year dataset. Specifically, this study aims to:

1. Merge the five temporal datasets (1-year through 5-year horizons) into a unified dataset for comprehensive analysis
2. Conduct thorough exploratory data analysis to understand data characteristics, missing value patterns, and feature relationships

3. Implement data preprocessing techniques including missing value imputation and feature selection strategies
4. Develop and optimize two ensemble models: Random Forest and XGBoost classifiers
5. Evaluate model performance using both full feature sets and selected feature subsets
6. Compare model effectiveness through comprehensive evaluation metrics including ROC curves, AUC scores, and classification reports
7. Provide insights into the most predictive features and optimal modeling approaches for bankruptcy prediction

The choice of ensemble methods, specifically Random Forest and XGBoost, is motivated by their proven effectiveness in handling complex financial data, their ability to capture non-linear relationships, and their robustness to overfitting. These tree-based ensemble methods are particularly well-suited for bankruptcy prediction tasks due to their capacity to handle mixed data types, identify important features, and provide interpretable results.

This research contributes to the growing body of literature on bankruptcy prediction by providing a comprehensive comparison of ensemble methods across different feature selection strategies. The findings will offer practical insights for financial analysts and practitioners seeking to implement effective bankruptcy prediction systems in real-world applications.

The subsequent chapters present detailed exploratory data analysis, methodology and modeling approaches, and comprehensive results with discussions on model performance and practical implications for bankruptcy prediction in financial decision-making contexts.

Chapter 2

Data

This chapter presents a comprehensive exploratory data analysis of the bankruptcy dataset, encompassing data preprocessing, missing value analysis, feature selection strategies, and correlation assessments. The analysis provides crucial insights into data quality and feature relationships that inform subsequent modeling decisions.

The analysis begins with the integration of five separate ARFF files representing bankruptcy data across different time horizons (1-year through 5-year). The ARFF stands for Attribute-Relation File Format, which is a simple file format for storing data with attributes and relations. Each dataset was loaded with appropriate handling of missing values (denoted as '?' in the original files) and merged into a unified dataset containing 43,405 observations across 64 financial features plus temporal and target variables.

The consolidated dataset was structured with 64 financial features (X1 through X64), a binary bankruptcy indicator (0 for non-bankrupt, 1 for bankrupt), and a year variable indicating the prediction horizon.

This integration enables comprehensive analysis across multiple temporal contexts while maintaining data integrity and consistency. Details of the features are shown in Table 2.1.

Table 2.1: The set of features considered in classification process

ID	Description	ID	Description
X1	net profit / total assets	X33	operating expenses / short-term liabilities
X2	total liabilities / total assets	X34	operating expenses / total liabilities
X3	working capital / total assets	X35	profit on sales / total assets
X4	current assets / short-term liabilities	X36	total sales / total assets
X5	[(cash + short-term securities + receivables - short-term liabilities) / (operating expenses - depreciation)] * 365,	X37	(current assets - inventories) / long-term liabilities
X6	retained earnings / total assets	X38	constant capital / total assets
X7	EBIT / total assets	X39	profit on sales / sales
X8	book value of equity / total liabilities	X40	(current assets - inventory - receivables) / short-term liabilities
X9	sales / total assets	X41	total liabilities / ((profit on operating activities + depreciation) * (12/365))
X10	equity / total assets	X42	profit on operating activities / sales
X11	(gross profit + extraordinary items + financial expenses) / total assets	X43	rotation receivables + inventory turnover in days
X12	gross profit / short-term liabilities	X44	(receivables * 365) / sales
X13	(gross profit + depreciation) / sales	X45	net profit / inventory
X14	(gross profit + interest) / total assets	X46	(current assets - inventory) / short-term liabilities
X15	(total liabilities * 365) / (gross profit + depreciation)	X47	(inventory * 365) / cost of products sold
X16	(gross profit + depreciation) / total liabilities	X48	EBITDA (profit on operating activities - depreciation) / total assets
X17	total assets / total liabilities	X49	EBITDA (profit on operating activities - depreciation) / sales
X18	gross profit / total assets	X50	current assets / total liabilities
X19	gross profit / sales	X51	short-term liabilities / total assets
X20	(inventory * 365) / sales	X52	(short-term liabilities * 365) / cost of products sold
X21	sales (n) / sales (n-1)	X53	equity / fixed assets
X22	profit on operating activities / total assets	X54	constant capital / fixed assets
X23	net profit / sales	X55	working capital
X24	gross profit (in 3 years) / total assets	X56	(sales - cost of products sold) / sales
X25	(equity - share capital) / total assets	X57	(current assets - inventory - short-term liabilities) / (sales - gross profit - depreciation)
X26	(net profit + depreciation) / total liabilities	X58	total costs / total sales
X27	profit on operating activities / financial expenses	X59	long-term liabilities / equity
X28	working capital / fixed assets	X60	sales / inventory
X29	logarithm of total assets	X61	sales / receivables
X30	(total liabilities - cash) / sales	X62	(short-term liabilities * 365) / sales
X31	(gross profit + interest) / sales	X63	sales / short-term liabilities
X32	(current liabilities * 365) / cost of products sold	X64	sales / fixed assets

A systematic analysis of missing values revealed significant data quality challenges requiring careful attention. The missing value distribution analysis identified substantial missingness across multiple features, with varying degrees of data completeness. Figure 2.1 shows the percentage distribution of missing values across financial features. The missing value assessment revealed that feature X37, representing the ratio of (current assets - inventories) to long-term liabilities, exhibited the highest percentage of missing values at approximately 40% of all observations. This substantial missingness, combined with the availability of related financial ratio variables, led to the decision to remove X37 from subsequent analyses.

Features X21 and X27 also demonstrated notable missing value percentages but were retained for analysis due to their potential predictive value and the feasibility of reliable imputation. The missing value visualization clearly illustrated the heterogeneous distribution of missingness across features, providing guidance for preprocessing decisions.

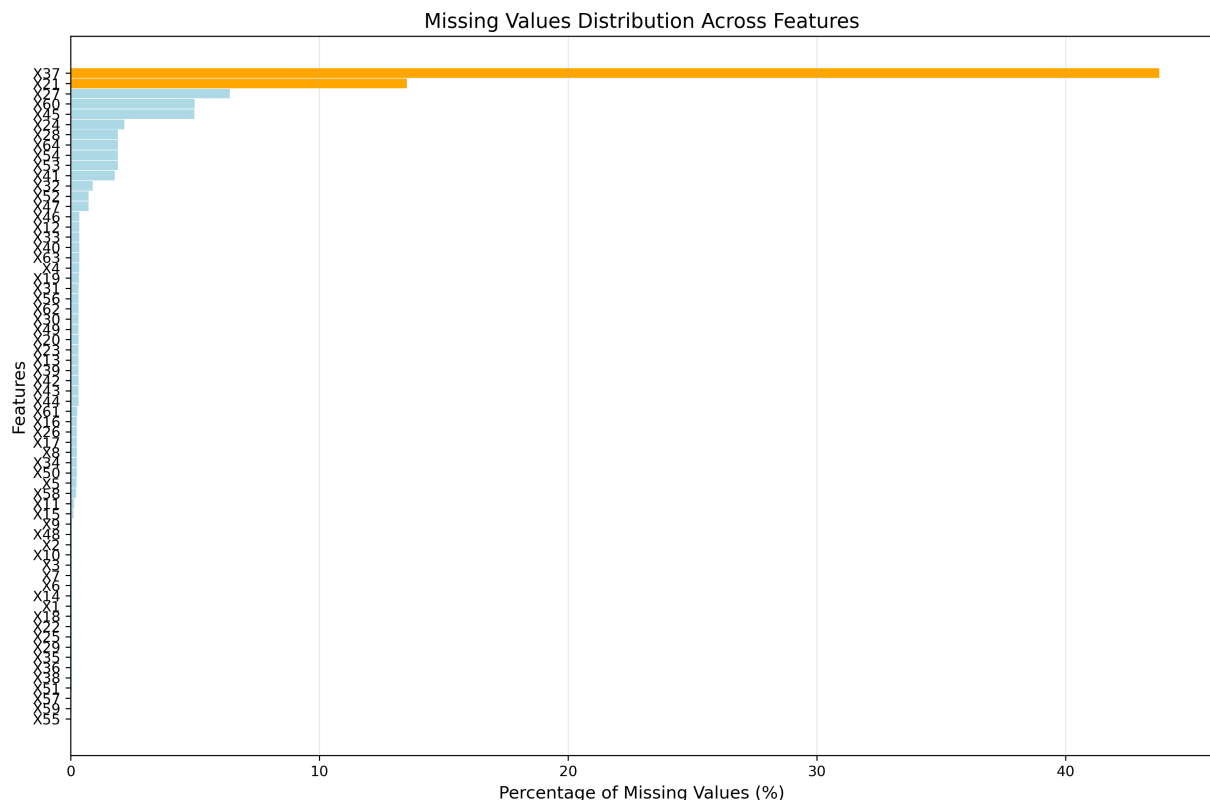


Figure 2.1: Distribution of Missing Values Across Financial Features

To address remaining missing values, K-Nearest Neighbors (KNN) imputation was implemented with $k=5$ neighbors and uniform weighting. This approach was

selected due to its ability to preserve local data structures and relationships while providing robust estimates for missing values based on similar observations.

The KNN imputation process considered the similarity between observations across all available features, ensuring that imputed values reflect the underlying data patterns. The choice of $k=5$ represents a balance between sufficient neighborhood information and computational efficiency, while uniform weighting ensures equal contribution from all neighbors in the imputation process.

Analysis of bankruptcy rates across different time horizons revealed interesting temporal patterns. The distribution of bankruptcy percentages by year showed varying rates of financial distress, with certain prediction horizons exhibiting higher bankruptcy frequencies. This temporal variation underscores the importance of considering time-dependent factors in bankruptcy prediction modeling. Figure 2.2 shows the bankruptcy rate in target variable distributed by years. The bankruptcy rate analysis demonstrated increasing bankruptcy rate over the years, however, the rate change is not significantly different across different years with values ranging from around 4% to 7% per annum.

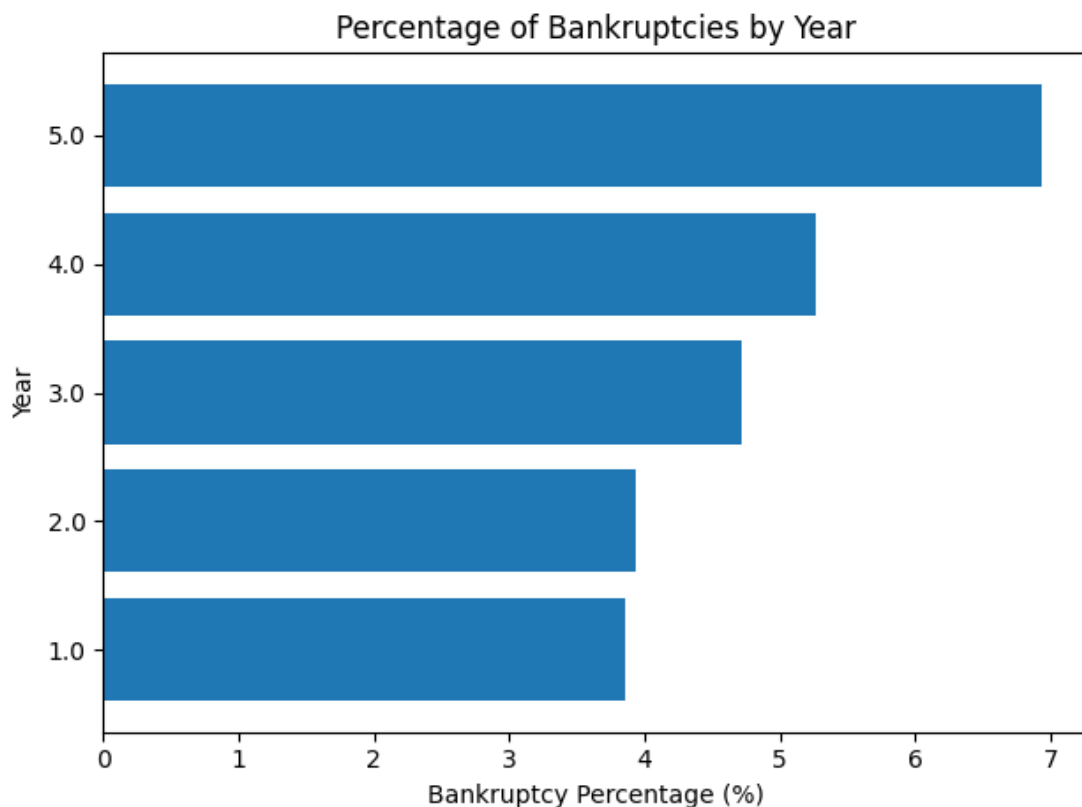


Figure 2.2: Bankruptcy Rate Distribution by Years

Comprehensive correlation analysis was conducted to identify relationships between financial features and inform feature selection strategies. The correlation matrix analysis revealed numerous strong positive correlations ($R > 0.95$) between feature pairs, indicating potential redundancy in the feature set.

The top 20 strongest correlation pairs were identified, with several correlations approaching perfect linear relationships. Notable examples include:

- X7 and X14 ($R = 1.000$)
- X4 and X46 ($R = 0.999$)
- X20 and X56 ($R = 0.999$)
- X8 and X17 ($R = 0.999$)
- X19 and X23 ($R = 0.999$)

These strong correlations suggest substantial feature redundancy and support the implementation of feature selection strategies to improve model efficiency and interpretability. The top 10 strongest correlation pairs are shown in Table 2.2.

Table 2.2: Top 10 Strongest Cross-Correlations Between Financial Features

Feature Pair	Correlation Coefficient	Relationship Strength
X7 - X14	1.000	Perfect
X4 - X46	0.999	Near Perfect
X20 - X56	0.999	Near Perfect
X8 - X17	0.999	Near Perfect
X19 - X23	0.999	Near Perfect
X19 - X31	0.999	Near Perfect
X43 - X44	0.999	Near Perfect
X10 - X38	0.999	Near Perfect
X53 - X54	0.999	Near Perfect
X23 - X31	0.998	Near Perfect

To address feature redundancy systematically, hierarchical clustering was applied to the correlation matrix using average linkage methodology. This approach groups features based on their correlation patterns, enabling the identification of feature clusters representing similar financial concepts.

The clustering analysis demonstrated in Figure 2.3 resulted in 8 distinct clusters, with cluster 8 identified as the most populous, containing 17 features. This

cluster represents a cohesive group of financial indicators sharing similar correlation patterns and underlying financial relationships.

From the top cluster (Figure 2.4), features X51 and X54 were removed due to their high correlation with other cluster members, resulting in a selected feature set of 15 variables. This feature selection strategy maintains representative coverage of financial concepts while reducing redundancy and computational complexity.

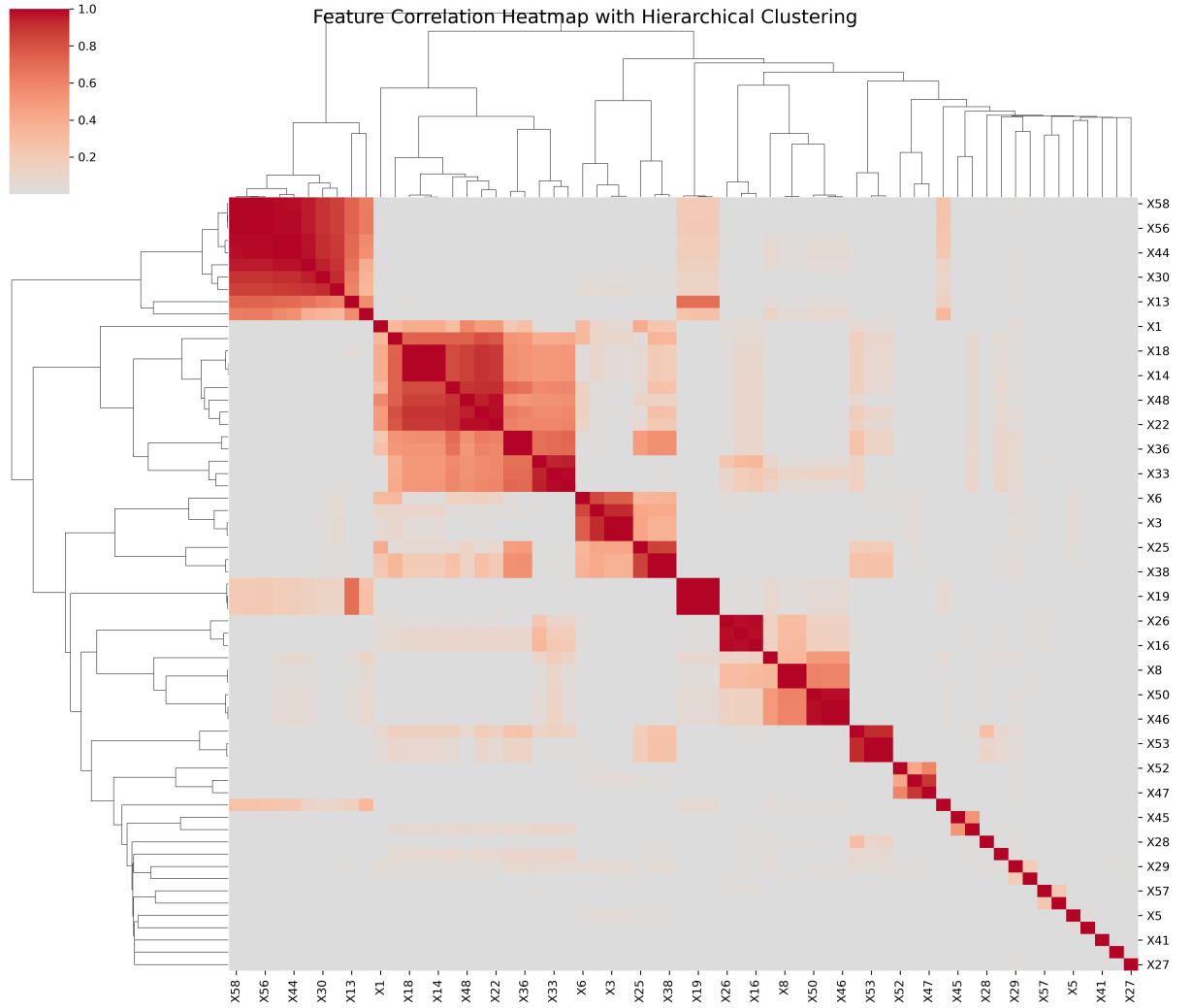


Figure 2.3: Hierarchical Clustering of Financial Features Based on Correlation Matrix. The clustermap shows the dendrogram and correlation heatmap, revealing 8 distinct clusters with cluster 8 being the most populous.

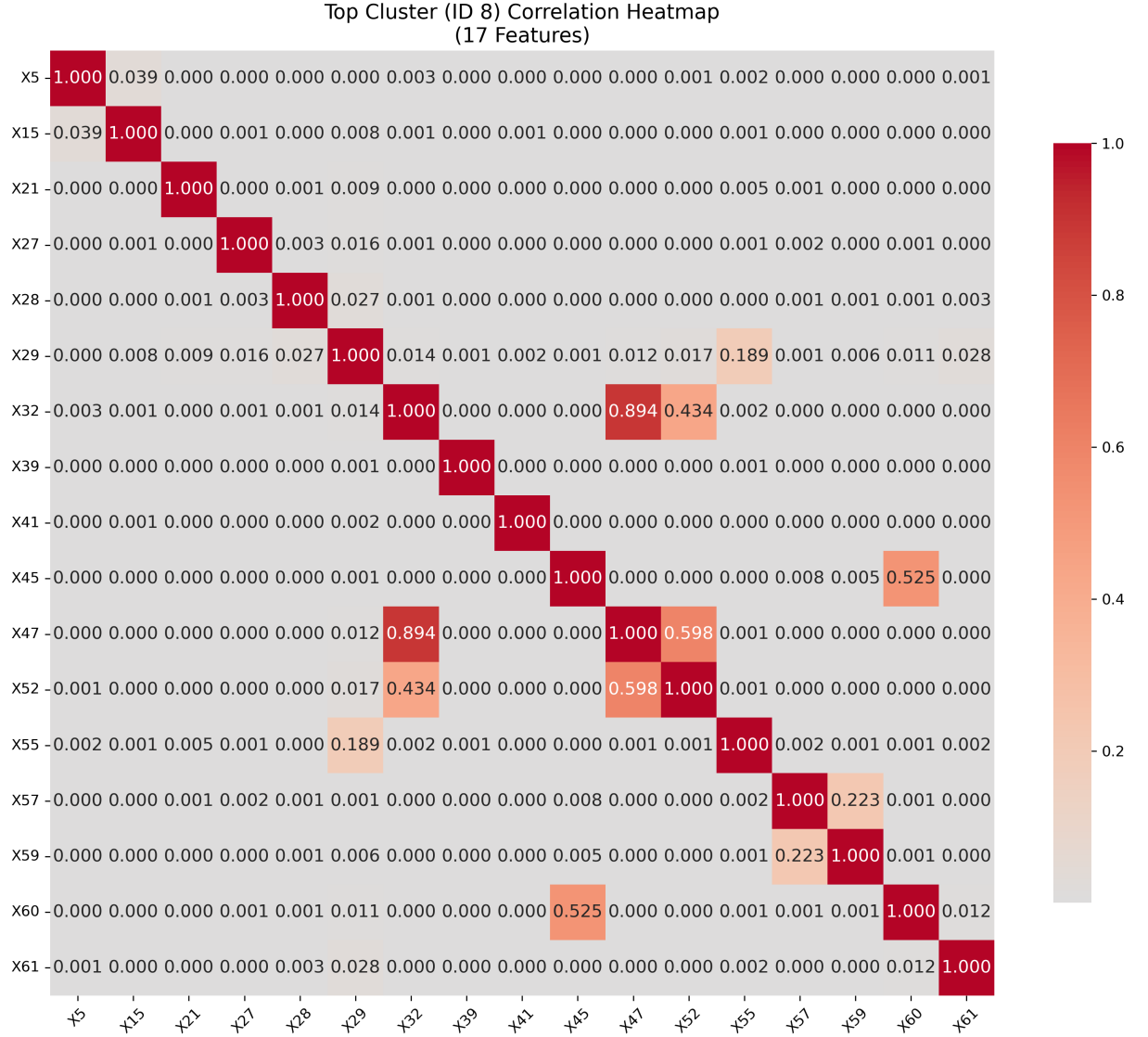


Figure 2.4: Correlation Heatmap of Top Cluster Features. This detailed view shows the correlation structure within the most populous cluster (cluster 8), containing 17 features before reduction.

To validate the feature selection strategy and assess multicollinearity, Variance Inflation Factor (VIF) analysis was conducted on the selected feature subset. The VIF analysis provides quantitative assessment of multicollinearity, with values below 5 generally indicating acceptable levels of feature independence.

The VIF analysis of the 15 selected features (Table 2.3) revealed satisfactory results, with all features exhibiting VIF values well below the commonly accepted

threshold of 5. The VIF values ranged from 1.000 to 2.694, indicating minimal multicollinearity concerns in the selected feature set.

Table 2.3: Variance Inflation Factor Analysis for Selected Features

Feature	VIF Value	Multicollinearity Level
X44	1.038	Very Low
X14	1.894	Low
X36	2.694	Moderate
X3	1.516	Low
X25	2.433	Moderate
X31	1.038	Very Low
X40	1.141	Very Low
X17	1.130	Very Low
X64	1.090	Very Low
X45	1.000	Minimal
X61	1.017	Very Low
X55	1.000	Minimal
X59	1.000	Minimal
X5	1.003	Very Low
X15	1.002	Very Low

The comprehensive exploratory data analysis established a solid foundation for subsequent modeling efforts. Key preprocessing achievements include:

1. Successful integration of multi-year bankruptcy data into a unified dataset
2. Systematic identification and handling of missing values through targeted removal and KNN imputation
3. Implementation of correlation-based feature selection reducing the feature space from 64 to 15 variables
4. Validation of feature selection through VIF analysis confirming minimal multicollinearity
5. Comprehensive understanding of bankruptcy distribution patterns across temporal horizons

The resulting preprocessed dataset provides both full feature (63 features after X37 removal) and selected feature (15 features) versions, enabling comparative analysis of model performance across different feature complexity levels. This

dual approach facilitates assessment of the trade-offs between model complexity and predictive performance in bankruptcy prediction applications.

Chapter 3

Modeling Methodology and Results

This chapter presents the implementation and evaluation of ensemble machine learning models for bankruptcy prediction. Two prominent ensemble methods (Bagging and Boosting style), Random Forest and XGBoost, were applied to both full and selected feature datasets, resulting in four distinct modeling approaches for comprehensive performance comparison.

Random Forest represents a robust ensemble method that combines multiple decision trees through bootstrap aggregating (bagging). Each tree in the forest is trained on a different bootstrap sample of the data, and predictions are made through majority voting for classification tasks. The method incorporates randomness at both the observation level (through bootstrap sampling) and feature level (through random feature selection at each split), reducing overfitting and improving generalization performance. Key advantages of Random Forest for bankruptcy prediction include its ability to handle mixed data types, resistance to overfitting, implicit feature selection capabilities, and provision of feature importance measures. The method's ensemble nature makes it particularly suitable for financial data where relationships may be complex and non-linear.

XGBoost (Extreme Gradient Boosting) on the other hand, represents an advanced implementation of gradient boosting that sequentially builds decision trees to correct errors from previous iterations. Unlike Random Forest's parallel tree construction, XGBoost employs a sequential approach where each tree learns from the residuals of its predecessors, resulting in progressively improved predictions. XGBoost incorporates several sophisticated features including regularization terms to prevent overfitting, advanced tree pruning algorithms, and optimized computational efficiency. These characteristics make it particularly effective for complex financial prediction tasks where subtle patterns and feature interactions are crucial for accurate classification.

Four distinct models were developed to enable comprehensive performance assessment:

1. Random Forest with Full Features (RF_Full): Utilizing all 63 features after X37 removal
2. Random Forest with Selected Features (RF_Selected): Utilizing the 15 clustered features
3. XGBoost with Full Features (XGB_Full): Utilizing all 63 features after X37 removal
4. XGBoost with Selected Features (XGB_Selected): Utilizing the 15 clustered features

Each model employed an 80-20 train-test split with consistent random state (123) to ensure reproducible results and fair comparison across modeling approaches.

Hyperparameter Optimization

Random Forest Hyperparameter Tuning

Comprehensive hyperparameter optimization was conducted using 5-fold cross-validation with GridSearchCV. The parameter search space included:

- Criterion: ['gini', 'entropy']
- Max features: [5, 10, 15, 20]
- Max depth: [None, 1, 3, 5, 7, 9]
- Min samples split: [8, 6, 4]
- Min samples leaf: [4, 2, 1]

Detailed results from hyperparameter tuning for Random Forest models are shown in Table 3.1.

Table 3.1: Optimal Hyperparameters for Random Forest Models

Hyperparameter	RF_Full	RF_Selected
Criterion	entropy	entropy
Max Depth	None	7
Max Features	20	10
Min Samples Leaf	2	2
Min Samples Split	8	8
CV Score	0.9582	0.9514

XGBoost Hyperparameter Tuning

XGBoost hyperparameter optimization employed a custom random search approach with 30 iterations, optimizing AUC scores through 5-fold cross-validation. The search space included:

- Max depth: [2, 4, 6, 8, 10, 12, 15, 20]
- Subsample: [0.5, 1.0]
- Colsample bytree: [0.5, 1.0]
- Colsample bylevel: [0.5, 1.0]
- Colsample bynode: [0.5, 1.0]

The optimization process identified optimal configurations for both full and selected feature models, with the full feature model achieving superior cross-validation AUC scores compared to the selected feature variant. Detailed results from hyperparameter tuning for XGBoost models are shown in Table 3.2.

Table 3.2: XGBoost Hyperparameter Search Results

Parameter	Search Range	Optimal Configuration
Max Depth	[2, 4, 6, 8, 10, 12, 15, 20]	15 (Full), 8 (Selected)
Subsample	[0.5, 1.0]	0.75
Colsample Bytree	[0.5, 1.0]	0.8-0.9
Colsample Bylevel	[0.5, 1.0]	0.6-0.8
Colsample Bynode	[0.5, 1.0]	0.7-0.9
Learning Rate	Fixed	0.1
N Estimators	Fixed	100

Model Performance Results

The performance results for all four models are shown in Table 3.3. We can see that XGBoost models generally perform slightly better than Random Forest models, with XGBoost_Full achieving the highest overall accuracy (96.9%) and XGBoost_Selected achieving the highest bankruptcy precision (0.89). With the selected feature models, the performance is slightly lower than the full feature models, but the difference is not significant. The reason is that the selected feature models are not able to capture the same level of information as the full feature

models. Both Random Forest and XGBoost models are rooted in tree-based model, which does not require feature scaling. Therefore, it is not necessary to scale the features for these models as shown in the EDA section.

Table 3.3: Comprehensive Model Performance Comparison

Metric	RF_Full	RF_Selected	XGB_Full	XGB_Selected
Overall Accuracy	95.5%	94.8%	96.9%	94.9%
Bankruptcy Precision	0.56	0.05	0.89	0.17
Bankruptcy Recall	0.07	0.01	0.36	0.03
Bankruptcy F1-score	0.12	0.01	0.52	0.05
Macro Avg F1-score	0.55	0.49	0.75	0.51
Weighted Avg F1-score	0.94	0.93	0.96	0.93

Receiver Operating Characteristic (ROC) analysis serves as the primary evaluation framework for assessing model discriminative ability across all decision thresholds. ROC curves plot the True Positive Rate (Sensitivity) against the False Positive Rate (1-Specificity), providing a threshold-independent assessment of classification performance. The Area Under the Curve (AUC) summarizes this performance into a single metric, where values closer to 1.0 indicate superior discriminative ability.

The ROC analysis revealed substantial performance differences across the four modeling approaches. Figure 3.1 presents the complete ROC curves for all models, demonstrating clear performance hierarchies and the impact of both algorithmic choice and feature selection strategies.

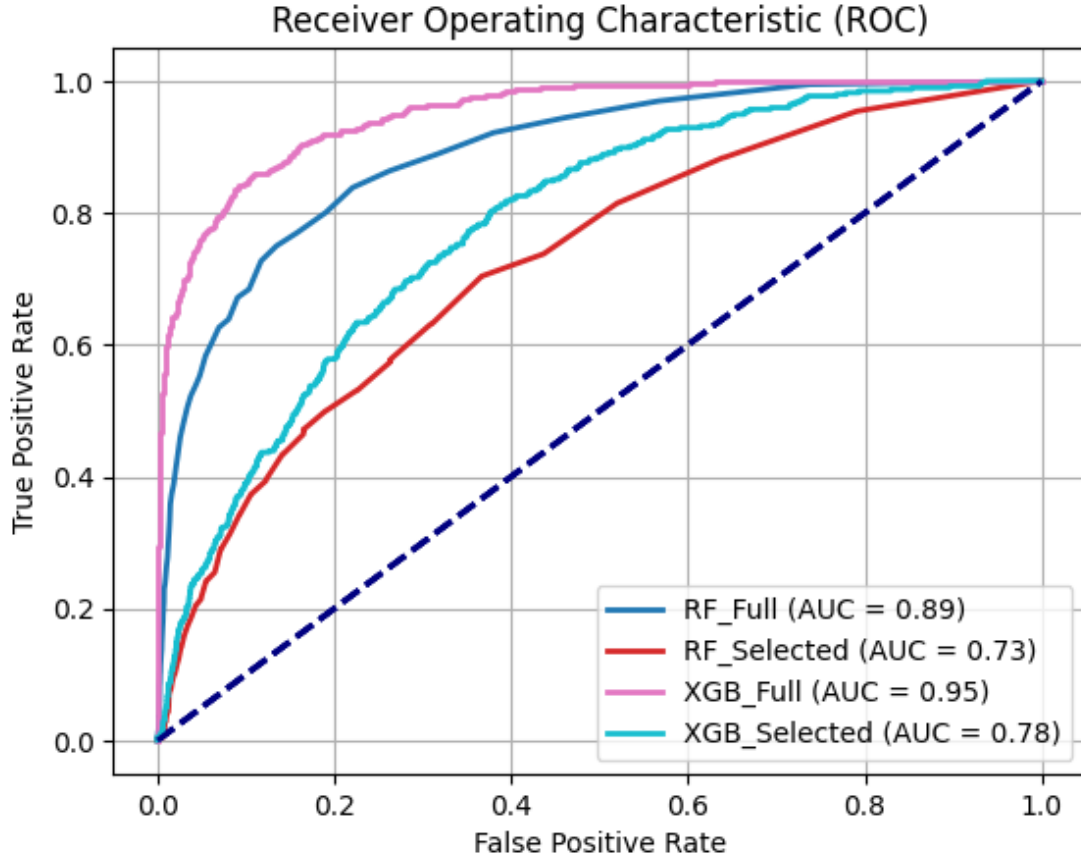


Figure 3.1: ROC Curves Comparison for All Four Models

Model-Specific ROC Performance:

- **XGB_Full (AUC = 0.95):** Exhibits exceptional discriminative ability with the curve positioned closest to the top-left corner, indicating optimal sensitivity-specificity balance across all threshold values.
- **RF_Full (AUC = 0.89):** Demonstrates strong performance but with noticeable deviation from the optimal curve, particularly in the high-sensitivity region crucial for bankruptcy detection.
- **XGB_Selected (AUC = 0.78):** Shows moderate discriminative ability with significant performance degradation compared to the full feature variant, highlighting the importance of comprehensive feature sets.
- **RF_Selected (AUC = 0.73):** Exhibits the weakest performance with substantial deviation from optimal classification, indicating limitations in both algorithmic approach and feature representation.

Feature Importance Analysis

Feature importance analysis revealed key financial indicators driving bankruptcy predictions. Both Random Forest and XGBoost models have option to calculate feature importance. The feature importance is calculated by the mean decrease in impurity (MDI) for Random Forest and gain for XGBoost. The top 10 most important features for Random Forest and XGBoost Full and Selected models are shown in Figure 3.2 and Figure 3.3 respectively.

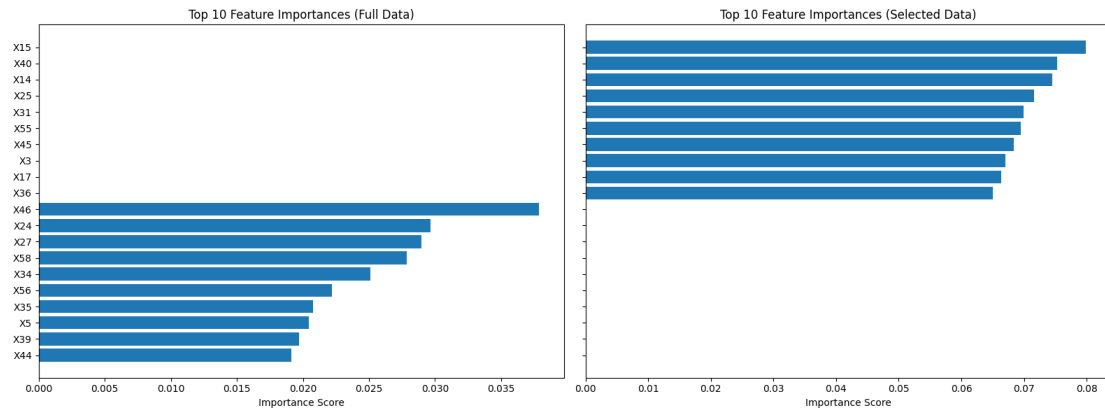


Figure 3.2: Feature Importance Analysis for Random Forest Models

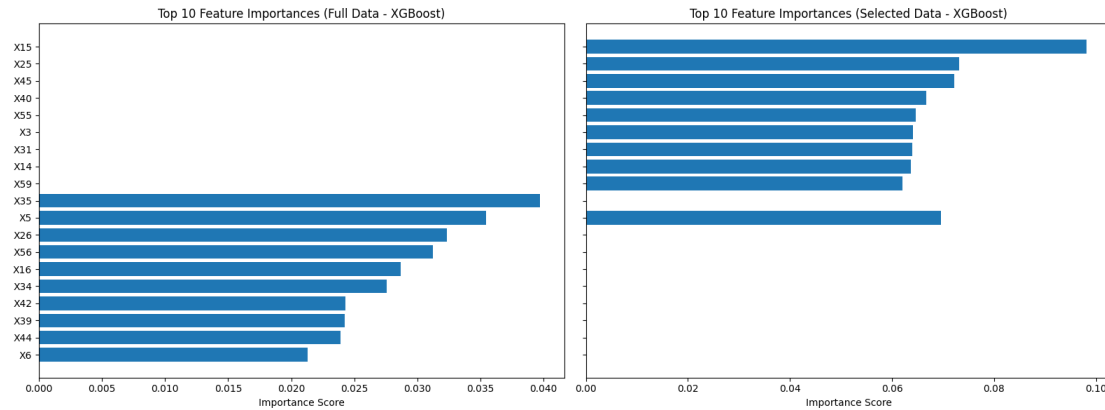


Figure 3.3: Feature Importance Analysis for XGBoost Models

From the feature importance analysis, we can see that there exists a significant difference between the top 10 most important features for Random Forest and XGBoost models for both full and selected feature models. The reason is that for Selected features we already removed the highly correlated features, which might

be in the top features from Full model. Among 10 important features, 5 features are the common for both Random Forest and XGBoost models with Full data, which are X35, X5, X56, X39, X44. This might be explained by the fact that the Random Forest and XGBoost models are different in the way they calculate feature importance. For Selected data, there are more common features are X15, X40, X14, X25, X31, X55, X45, X3.

Top 5 common Important Features (XGB_Full and RF_Full):

1. X35: Profit on sales / total assets
2. X5: $[(\text{cash} + \text{short-term securities} + \text{receivables} - \text{short-term liabilities}) / (\text{operating expenses} - \text{depreciation})] * 365$
3. X56: $(\text{sales} - \text{cost of products sold}) / \text{sales}$
4. X39: Profit on sales / sales
5. X44: $(\text{receivables} * 365) / \text{sales}$

All models underwent rigorous 5-fold cross-validation during hyperparameter optimization, ensuring robust performance estimates and reducing overfitting risks. The cross-validation results demonstrated consistent performance across folds, validating the reliability of the modeling approaches.

Early stopping mechanisms were implemented in XGBoost training to prevent overfitting and optimize model complexity. The training process monitored both training and validation AUC scores, terminating when validation performance plateaued to ensure optimal generalization.

Training times varied significantly across models, with Random Forest models generally requiring longer training periods due to the larger ensemble sizes (100 trees vs. optimized boosting rounds in XGBoost). The selected feature models demonstrated reduced computational requirements while maintaining reasonable performance levels, highlighting the trade-off between feature complexity and computational efficiency.

Chapter 4

Discussion and Conclusions

Discussions

The comprehensive evaluation of four distinct modeling approaches reveals clear performance hierarchies and trade-offs between different strategies. The analysis encompasses multiple evaluation metrics including overall accuracy, class-specific performance measures, and discriminative ability through ROC-AUC analysis.

Based on the comprehensive evaluation metrics, the models rank as follows in terms of overall effectiveness:

1. **XGBoost Full Features (XGB_Full)**: Superior performance across all key metrics
2. **Random Forest Full Features (RF_Full)**: Strong performance with some limitations in minority class detection
3. **XGBoost Selected Features (XGB_Selected)**: Moderate performance with computational efficiency benefits
4. **Random Forest Selected Features (RF_Selected)**: Lowest performance across most metrics

The detailed classification reports reveal important insights into model behavior across different classes:

For Minority Class (Bankruptcy) Detection: XGB_Full achieves the highest bankruptcy detection capability with 36% recall and 89% precision, representing the best balance between sensitivity and specificity for bankruptcy identification. This performance is particularly valuable in practical applications where accurate bankruptcy detection is prioritized over overall accuracy.

Random Forest models demonstrate poor minority class detection, with RF_Full achieving only 7% recall and RF_Selected achieving merely 1% recall for bankruptcy cases. This limitation significantly reduces their practical utility for bankruptcy prediction applications.

For Majority Class (Non-Bankruptcy) Performance: All models achieve excellent non-bankruptcy classification performance, with precision and recall values exceeding 95%. This consistent performance across models reflects the class imbalance in the dataset and the relative ease of identifying financially stable companies.

Limitations and Future Research Directions

The severe class imbalance in the dataset (approximately 95% non-bankrupt cases) presents ongoing challenges that merit further investigation. Future research should explore advanced sampling techniques, cost-sensitive learning approaches, and threshold optimization strategies to improve minority class detection.

While this study successfully integrated multi-year data, the temporal aspects of bankruptcy prediction warrant deeper investigation. Future research could explore time-series modeling approaches and dynamic feature importance evolution across prediction horizons.

The current analysis relies on a single dataset source. External validation using independent bankruptcy datasets would strengthen confidence in the modeling recommendations and assess generalizability across different economic contexts and time periods.

Conclusions

This comprehensive study of ensemble methods for bankruptcy prediction yields several definitive conclusions:

Algorithmic Superiority: XGBoost demonstrates clear superiority over Random Forest for bankruptcy prediction tasks, with substantial improvements in discriminative ability and minority class detection.

Feature Completeness Importance: Comprehensive feature sets significantly outperform reduced feature sets, challenging conventional wisdom about feature selection for tree-based ensemble methods.

Performance Trade-offs: While computational efficiency improvements are achievable through feature selection, the performance costs are prohibitive for applications prioritizing prediction accuracy.

Practical Recommendations: XGBoost with full features represents the optimal approach for bankruptcy prediction, offering the best balance of discrim-

inative ability, class-specific performance, and overall accuracy.

These findings provide clear guidance for practitioners implementing bankruptcy prediction systems and establish a robust foundation for future research in financial distress modeling. The demonstrated effectiveness of XGBoost with comprehensive feature sets offers practical value for financial institutions, investment firms, and regulatory agencies seeking reliable bankruptcy prediction capabilities.